

1 Basic Overview

The GUI system is a user-friendly interface designed to control an underwater camera lighting and capture setup. This software provides researchers with an easy way to adjust the Aquorea Mk3 LED lights (which function as both a constant lamp and a strobe flash) and to trigger a connected Sony camera for underwater observations. The GUI is intended for non-technical users – you can perform all key actions with simple buttons and sliders, without needing to write any code or memorize complex commands.

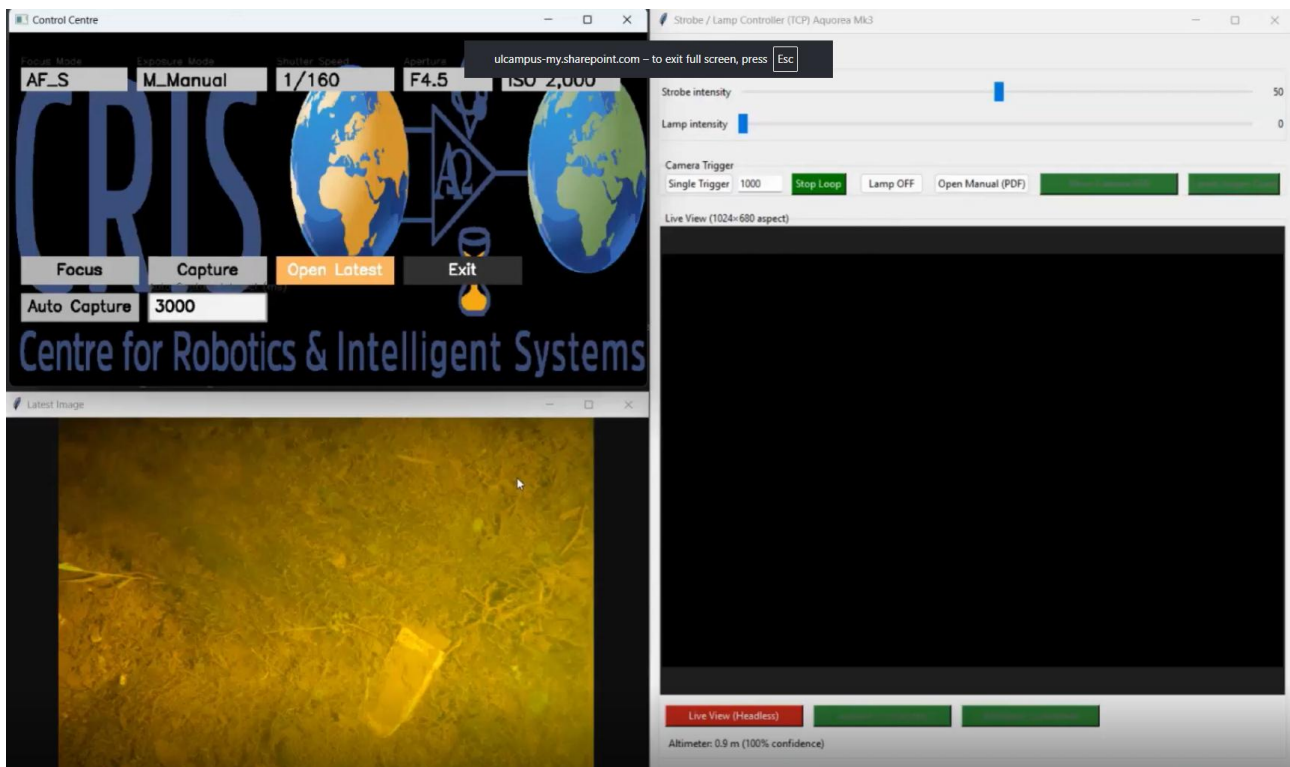


Figure 1: GUI

Purpose and Capabilities: The GUI allows you to manage lighting intensity, take photos (either one at a time or in a continuous loop), view live video from the camera, and monitor an altimeter (distance sensor) reading. It runs on a PC and communicates with the underwater equipment over a network connection. *(Technical note: The system uses TCP/IP for surface-to-underwater communication and an RS-485 link underwater – you do not need to manage these details, but it ensures commands from the GUI reliably reach the subsea lights and camera controller.)*

In summary, this GUI acts as the central control panel for your underwater camera system. With it, you can easily adjust the brightness of lights, capture images for research, and see real-time video – all through a clear, point-and-click interface.

2 Prerequisites (Sony ILX)

1. Make sure the Sony ILX-LR1 is powered on and connected via USB.
2. On the camera:
 - Go to Setup → USB Connection → PC Remote Function
 - Make sure Authentication Setting = Off



3. In the Remote Function menu, select Pairing and ensure the PC is properly paired.

3 Common Issues and Fixes

"I don't see the camera"

- Verify the USB cable and camera power.
- On the camera, check:
 - Setup → USB Connection → Authentication Setting is set to Off.

"Connection is refused"

- On the camera, go to PC Remote Function → Pairing
- Perform pairing again and issue should be resolved.

Note: The Live View panel and these connection buttons give you real-time feedback on system status. Generally, **green** indicates an active/connected state (for example, the button may turn green or disable itself when connected), and **red** indicates a stopped or disconnected state (the button is clickable and shown in red when a connection is needed). The GUI handles all this automatically on startup and lets you know if there is a connection issue.